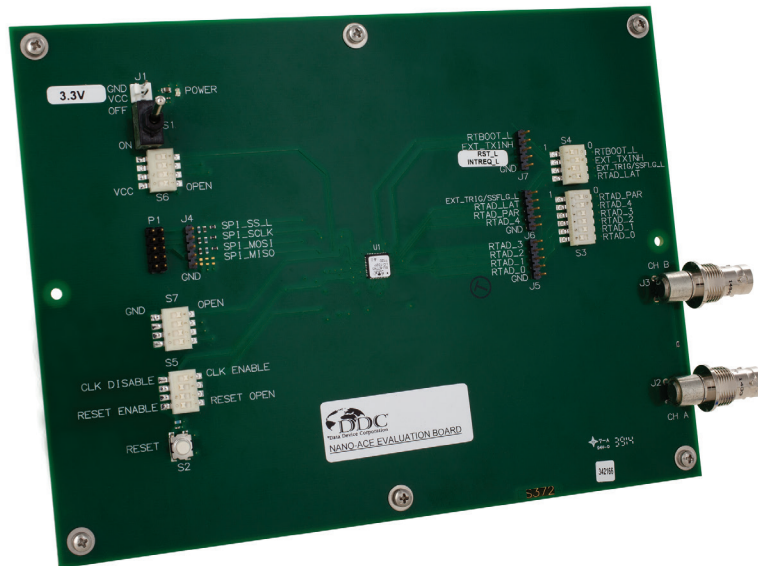


Nano Ace[®] Development Kit



User's Guide

Model: BU-67743EC / BU-67753EC / BU-67833EC



Nano-ACE is the world's smallest and lowest power complete, MIL-STD-1553 Remote Terminal/Monitor, enabling the development of more compact and higher density boards.

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- Simple Systems
- Radios/Modems
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NANO-ACE DEVELOPMENT KIT

USER'S GUIDE

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RECORD OF CHANGE

Revision	Date	Pages	Description
Rev A	Sept 2019	All	Initial release
Rev B	June 2020	All	Update to add BU-67833EC and BU-67753EC

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1 PREFACE

This manual uses typographical conventions to assist the reader in understanding the content. This section will define the text formatting used in the rest of the manual.

1.1 Text Usage

- **BOLD**—indicates important information.
- `Courier New`—indicates code examples.
- `<...>` - indicates user-entered text or commands.

1.2 Special Handling and Cautions

The Nano-ACE uses state-of-the-art components, and proper care should be used to ensure that the device will not be damaged by Electrostatic Discharge (ESD), physical shock, or improper power surges and that precautions are taken to avoid electrocution.



Warnings: Turn off power to the computer hardware and unplug from wall.

NEVER insert or remove card with power turned on.

Ensure that standard ESD precautions are followed. As a minimum, one hand should be grounded to the power supply in order to equalize the static potential.

Do not store disks in environments exposed to excessive heat, magnetic fields or radiation.

1.3 Trademarks

All trademarks are the property of their respective owners.

1.4 What is included in this manual?

This manual contains a complete description of hardware installation and use.

1.5 Technical Support

In the event that problems arise beyond the scope of this manual, you can contact DDC by the following:

US Toll Free Technical Support:
1-800-DDC-5757, ext. 7771

Outside of the US Technical Support:
1-631-567-5600, ext. 7771

Fax:
1-631-567-5758 to the attention of DATA BUS Applications

DDC Website:
www.ddc-web.com/ContactUs/TechSupport.aspx

Please note that the latest revisions of Software and Documentation are available for download at DDC's Web Site, www.ddc-web.com.

2 OVERVIEW

The Nano-ACE® is a complete and compact solution to MIL-STD-1553 applications. With a footprint of just 7mm by 7mm, it is the world's smallest terminal. The device is powered entirely by +3.3 volts.

Ideal for extended temperature range applications where PC board space is at a premium, the Nano-ACE is available in a 48-pad QFN package, and is rated for -55°C to +125°C operation at base.

The Nano-ACE is software and architecturally compatible with DDC's Enhanced Mini-ACE series of devices. It integrates dual transceivers, protocol engine and 4K or 32K words of internal RAM.

The Nano-ACE's Serial Peripheral Interface (SPI) allows direct connection with little or no glue logic and minimal pin usage to a variety of Microcontrollers, DSPs, and FPGAs.

The Nano-ACE Evaluation Boards, provide access to the Nano-Ace through a SPI interface. The Development kit comes with the Nano-ACE Evaluation board, a Cheetah SPI to USB adapter, and power cable. The BU-67833EC Evaluation Board uses the BU-67833LC (BC/RT/M). The BU-67753EC Evaluation Board uses the BU-67753LC (RT/MT), and the BU-67743EC Evaluation board uses the BU-67743L8 (RT).

2.1 Features

- World's Smallest MIL-STD-1553 BC/RT/MT and RT/MT
- Small Package QFN (7mm x 7mm x 1mm)
- 50MHz 4-wire Serial Peripheral Interface (SPI) to Host
- Fully Compatible with Enhanced Mini-ACE® Software and Architecture
- Minimal Signal Count SPI Processor Interface
- +3.3 Volt Only
- 4K or 32K words of internal RAM
- 1mm Max Height
- Compliance with MIL-STD-1553A/B Notice 2, STANAG-3838, and MIL-STD-1760
- Highly Autonomous BC Architecture (BU-67833LC)

- Built-In Message Sequence Controller with 20-Instruction Set
- Flexible RT Buffering with Single, Double, and Circular Buffering
- Selective Message Monitor with Filtering
- 50% Rollover Interrupts for Stacks and Circular Buffers
- Software and Register Compatible to ACE and Enhanced Mini-ACE devices
- RT-Only Operation for Safety-Critical Applications
- Auto-Configuration Mode (BU-67833LC and BU-67753LC)
- Windows® XP / 7 / 8 / 10 ®

2.2 System Requirements

- USB 2.0 Port.
- Windows XP (32-Bit) / Vista/7/8/10 (32/64-Bit)

2.3 Mechanical Design

The Evaluation board card has several DIP switches and jumpers used to configure the clock and RT address options of the Nano-ACE. The Evaluation board also has a toggle switch for power, a ten pin connector for the SPI interface and a two pin connector to supply 3.3 Volts to the board. The board has two Trompeter connectors for access to the MIL-STD-1553 bus, one connector for Bus A, the other for Bus B.



Figure 1. BU-67833EC Nano-ACE Evaluation Board



Figure 2. BU-67433EC Nano-ACE Evaluation Board

3 INSTALLATION

Installing the Nano-ACE Evaluation board requires the installation of the Cheetah SPI to USB drivers along with the AceXtreme C SDK for Windows. Hardware installation is done after the drivers have been installed and will require the supplied USB A to USB B cable along with the supplied power adapter.

3.1 Software Installation

The Cheetah SPI to USB driver installation is done by TotalPhaseUSB-vX.YZ.exe file included in the Development Kit. Running the executable will start the installation of the driver.

3.1.1 Total Phase USB Driver

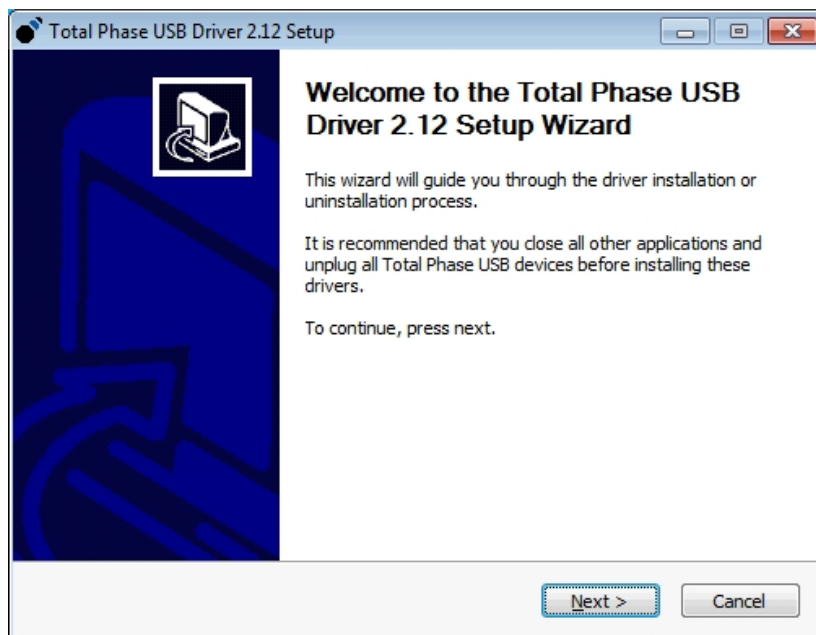


Figure 2. Total Phase USB Driver Installation

Clicking on the Next button will bring you to the Total Phase License agreement page.

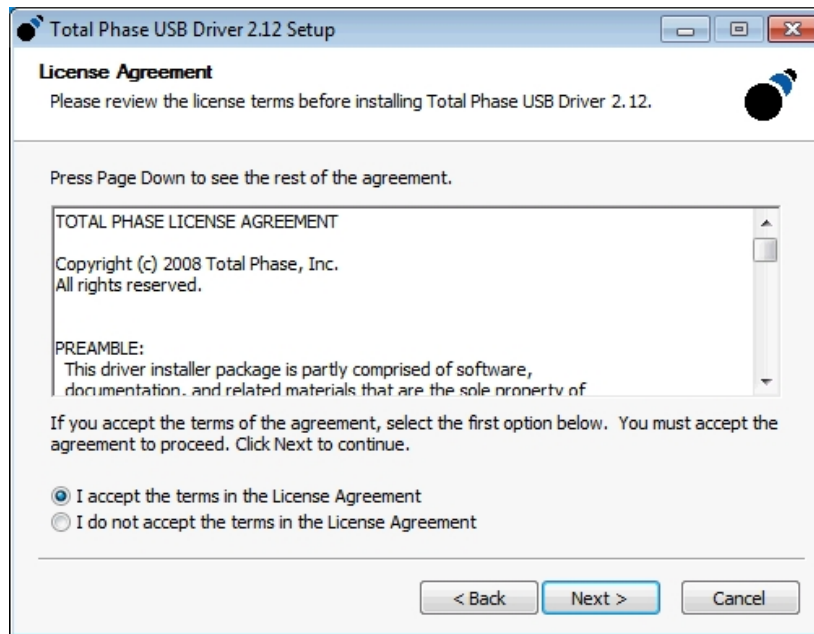


Figure 3. License Agreement

Once you have reviewed the license and clicked on the “I Accept the terms in the License Agreement” radio button, the “Next” button will be enabled allowing the user to move to the next step of the installation.

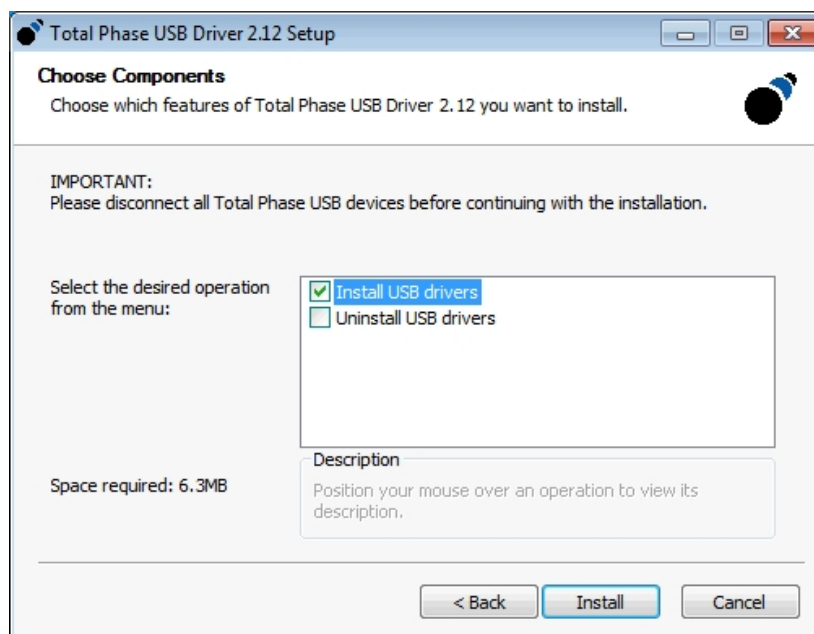


Figure 4. Installation options

After clicking the next button the user must select to either Install or Uninstall the USB drivers. At this point select the Install USB Drivers option and press the Install button to begin the driver installation.

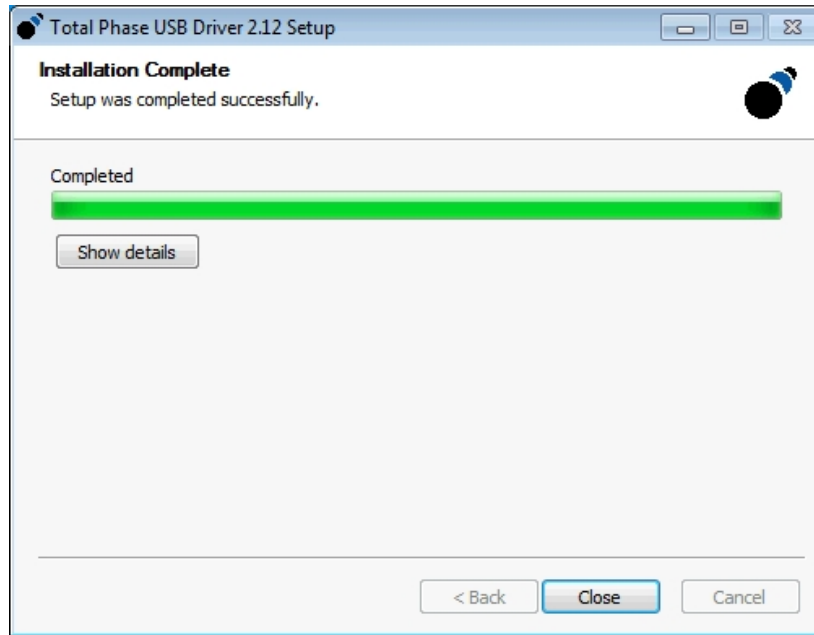


Figure 5. Installation Complete

Once the installation wizard shows the driver installation is completed, click the close button to exit the installer.

3.2 Hardware Installation

After the Cheetah USB drivers have been installed, the Cheetah SPI to USB interface and power adapter can be connected to the Evaluation board.

Before connecting power adapter cable, make sure the Power toggle switch (S1) is in the OFF position. The power cable is keyed so that it can only be connected one way as shown in Figure 6.

The 10 pin connector on the Cheetah can now be plugged into the P1 pin block on the Evaluation board also as show in Figure 6.



Figure 6. Cheetah connection

After the Cheetah has been connected you can plug in the USB cable into the Cheetah SPI to USB interface and to the host computer to complete the hardware installation. At this point, the power switch (S1) can be put into the ON position to power on the Nano-ACE and is ready for operation.

3.3 Hardware Configuration and Operation

The Nano-ACE Evaluation board uses the Cheetah SPI to USB interface to connect to a Windows Operating system. The Evaluation board does not require any jumpers or switches to set the Base address or interrupt values. The job of configuration for Plug-and-Play PCI is performed by the operating system.

4 ARCHITECTURE

4.1 Board Pinouts

This section delineates the user's pin outs for the card. The connectors described here are the two connectors on the I/O bracket and the few configuration jumpers on the board itself.

4.1.1 Power Toggle Switch

Power to the Nano-ACE is controlled through the toggle switch labeled as **S1** on the Evaluation board. Putting the switch into the "ON" position will provide the necessary power (3.3 Volts) to the Nano-ACE. Positioning the switch to the "OFF" position powers off the Nano-ACE. It should be noted the Cheetah SPI to USB interface is powered through the USB bus and not through the power adapter connected to the Evaluation board.

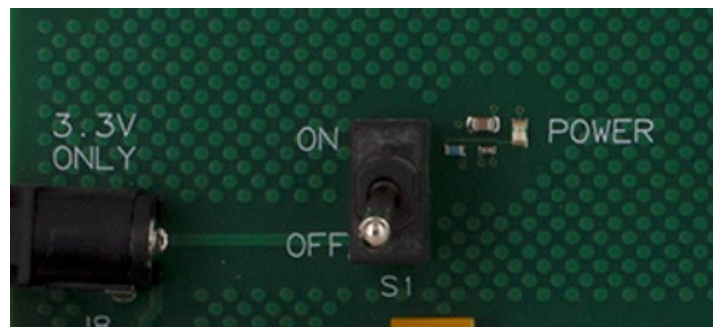


Figure 7. Power Toggle Switch

4.1.2 Reset Button

The reset button labeled **S2** can be used to cycle the power on the Nano-ACE resetting it to its default power-up state.



Figure 8. Reset button

4.1.3 DIP Switches

The Nano-Ace Evaluation board does contain several DIP switch blocks used to configure the Nano-ACE. The DIP Switches give the user control over setting the RT address, Transmit inhibit, the RT's subsystem flag bit, and the Nano-ACE clock.

4.1.3.1 RT Address Lines

The DIP switch block labeled **S3** is used to set the RT address and the parity bit. Odd parity is used for setting the RT address. For further information on setting the RT address please refer to the Nano-ACE data sheet.

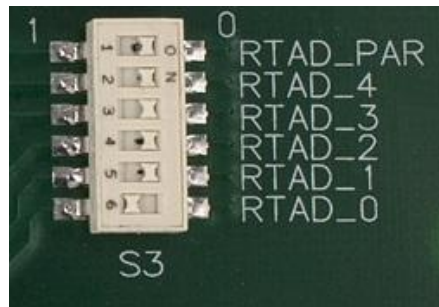


Figure 9. RT Address and Parity

4.1.3.2 RT Options

The **S4** DIP Switch block is used to configure the RT Address latch ping, the External Trigger / Sub System Flag and if the Nano-ACE will use RT Auto-boot. For further information on setting these options please refer to the Nano-ACE data sheet.

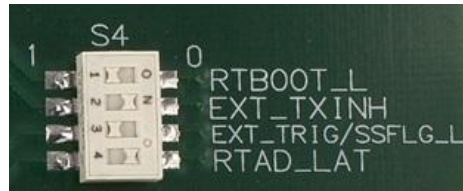


Figure 10. RT options

4.1.3.3 S5 DIP Switch

The S5 DIP Switch block is used to configure the clock options on the Nano-ACE. Enabling the Auto-Initialize feature in the Nano-ACE, and configuring the Flash Write protection of the EPPROM (U2 on the board) is also configured with the S5 DIP Switch block.

The EEPROM is used to Auto-Initialize the Nano-ACE for the BU-67753L and BU-67833L. The BU-67743L does not have the Auto-Initialization feature. Please see the Nano-Ace Data Sheet (DS-BU-67743L-C) for more information on the Auto-Initialization feature.

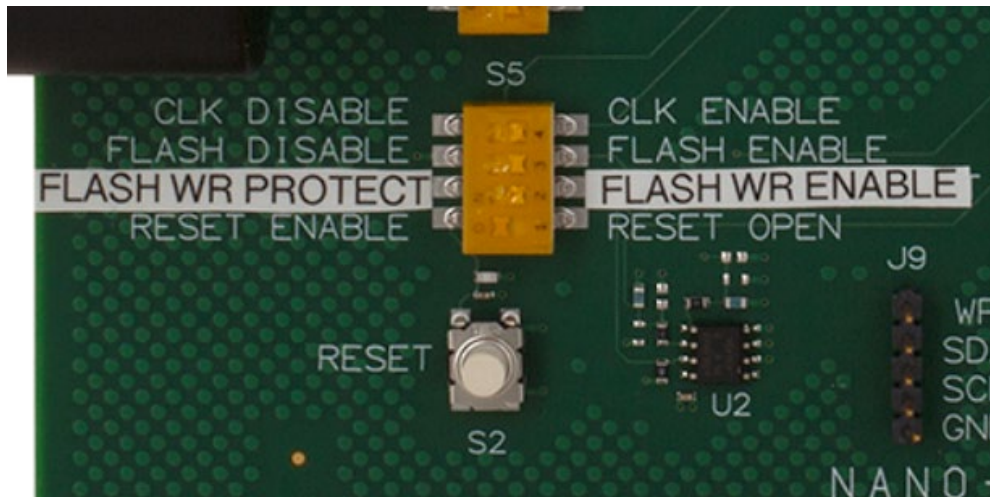


Figure 11. Clock DIP Switch

4.1.4 Test Points

The Evaluation board has several test points labeled J4 through J9. These test points are connected to the following SX blocks:

- J4 is the test point for the SPI interface used by the Cheetah.
- J5, J6, J7 pins are the test point for the S3 DIP Switch block (RT Address) S4 DIP Switch block (RT options).
- J9 pins are the tests points for the EEPROM (only available on BU-67883EC).

4.1.5 MIL-STD-1553 Connections

The J2 and J3 Trompeter connectors are used to connect the Nano-ACE to the MIL-STD-1553 bus. J2 is for Channel A, and J3 is for Channel B.

6 SOFTWARE

6.1 Overview

The card is supplied with the BU-69095S MIL-STD-1553 Nano-ACE C Software Development Kit (SDK) for Windows XP / 7 / 8 / 10. This software development kit includes a runtime library that provides the user with a hardware abstraction layer for the DDC 1553 hardware.

6.1.1 Windows XP / 7 / 8 / 10 - BU-69095S0

Software installation is extracting the zip file labeled BU69095S0_X_Y_Z.zip (where X_Y_Z is the version of the software package) to the C:\DDC folder. If the C:\DDC folder is not already created, please create the folder. If desired a different folder location may be used.

6.1.2 Contents of BU-69095S0

The BU-69095S0 currently contains three folders, a “docs”, “libraries” and “samples” folder. The “docs” contain files with relevant information regarding the release such as a readme and release notes file. The “libraries” folder contains the library source and project files for the Nano-ACE. The project file uses Microsoft Visual Studio 2017.

6.2 API

The BU-69095S0 Nano-Ace Software Development Kit (SDK) was written to interface with the Cheetah SPI to USB adapter. The source of the SDK is provided in the case a different SPI interface is being used. In this case, the SDK low level API calls will need to be modified to add support for the different SPI interface. The low level calls of the SDK are in the hal.c module and start with the prefix “naHal”.

For writing an application with the SDK, the upper level function calls will be used. These calls reside in the na.c module and start with the prefix “na”. The following section describes the available functions in the SDK for application development.

nalnit

Initializes the Nano-ACE

PROTOTYPE

```
#include "na.h"
```

```
int16_t nalnit();
```

PARAMETERS

None

DESCRIPTION

This function is used to initialize the Nano-ACE into its default state. This is the first function call necessary to initiate communication from the host system to the Nano-ACE as the SPI to USB connection is started.

RETURN VALUE

NA_SUCCESS
NA_FAILURE

The function completed without any errors.
There was an error communicating with the Nano-ACE.

naFree

This function is used to close the connection to the Nano-ACE

PROTOTYPE

```
#include "na.h"
```

```
int16_t naFree();
```

PARAMETERS

None

DESCRIPTION

This function releases any resources being used by the Nano-ACE and closes the communication handle between the Nano-ACE and the host CPU.

RETURN VALUE

NA_SUCCESS
NA_FAILURE

The function completed without any errors.
There was an error communicating with the Nano-ACE.

naConfigSPI

This function is used to configure the SPI interface options.

PROTOTYPE

```
#include "na.h"
```

```
int16_t naConfigSPI(uint8_t u8Polarity, uint8_t u8Phase, uint16_t u16FrequencyKHz);
```

PARAMETERS

u8Polarity	(input parameter) Sets the clock's idle state. Valid values: NA_CPOL_INACTIVE_LOW. NA_CPOL_INACTIVE_HIGH.
u8Phase	(input parameter) Configures the data sample, Valid values: NA_CPHA_1ST_EDGE. NA_CPHA_2ND_EDGE.
u16FrequencyKHz	(input parameter) SPI Communication Frequency which is dependent on the limitation of the SPI master device. Valid values: Up to 50 MHz max.

DESCRIPTION

This function is used to set the Clock Polarity, Clock Mode and Clock Frequency. The SPI standard defines four possible modes of operation relating to clock polarity (CPOL) and the clock phase (CPHA). The Nano-Ace supports mode 0 (CPOL= 0 & CPHA = 0) and mode 3 (CPOL =1 & CPHA =1).

RETURN VALUE

NA_SUCCESS	The function completed without any errors.
NA_FAILURE	There was an error communicating with the Nano-ACE.
NA_INVALID_ARGUMENT	One of the arguments passed into the function is incorrect.

naSlaveSelect

This function is used to configure the Slave Select lines.

PROTOTYPE

```
#include "na.h"
```

```
int16_t naSlaveSelect(uint8_t u8Enable, uint8_t u8Slaves);
```

PARAMETERS

u8Enable	(input parameter) Used to specify if the line(s) will be enabled or disabled. Valid values: NA_SS_ENABLE NA_SS_DISABLE
u8Slaves	(input parameter) Bit Field to determine which slave select line will be modified. Valid values: NA_SS1 NA_SS2 NA_SS3 NA_SS_ALL

DESCRIPTION

This function is used to enable or disable the Slave Select line.

RETURN VALUE

1, 2, 3	The number of the active slaves.
NA_FAILURE	There was an error communicating with the Nano-ACE.
NA_INVALID_ARGUMENT	One of the arguments passed into the function is incorrect.

naSendProtocol

Creates the message to be sent.

PROTOTYPE

```
#include "na.h"
```

```
int16_t naSendProtocol(uint8_t u8TransferType,
                      uint8_t u8ReadWrite,
                      uint8_t u8MemReg,
                      uint16_t u16Address,
                      uint16_t *pArray,
                      uint8_t u8WordCount);
```

PARAMETERS

u8TransferType	(input parameter) Specifies the transfer type. Valid values: NA_SIMPLE NA_SINGLE NA_BURST
u8ReadWrite	(input parameter) Configures the transfer as a read or write operation. Valid values: NA_READ NA_WRITE
u8MemReg	(input parameter) Defines either a memory or register transaction. Valid values: NA_MEMORY NA_REGISTER
u16Address	(input parameter) The Address of the Memory or Register.
pArray	(input / output parameter) Array containing the data to be written or read.
u8WordCount	(input parameter) Denotes the desired number of elements to be written or read. If the NA_BURST option is chosen for u8TransferType, then u8WordCount can be set from 0 to 255. 256 words will be read or written if 0 is entered for this parameter.

naSendProtocol (continued)

DESCRIPTION

This function is used to create the messages to be written or read from the Nano-ACE.

RETURN VALUE

1, 2, 3	The number of data words written to/read from the device.
NA_FAILURE	There was an error communicating with the Nano-ACE.
NA_INVALID_ARGUMENT	One of the arguments passed into the function is incorrect.
NA_NOT_CONFIGURED	The SPI interface was not configured properly.
NA_SLAVE_NOT_SELECTED	The Slave Select line was not configured.
NA_NULL_POINTER	The array was not configured properly.
NA_NOT_ALL_BYTES_SENT	Total number bytes transmitted were less than requested.
NA_PROTOCOL_MISMATCH	Invalid protocol.
NA_INVALID_ADDRESS	The Memory or Register Address is invalid.

naSendRaw

Creates the message to be sent in a raw format

PROTOTYPE

```
#include "na.h"
```

```
int16_t naSendRaw(uint8_t *pInputArray, uint8_t *pOutputArray, uint16_t u16ByteCount);
```

PARAMETERS

pInputArray	(input parameter) Array containing all raw SPI input bytes.
pOutputArray	(output parameter) Array containing all SPI output bytes.
u16ByteCount	(input parameter) Number of bytes to be transferred.

DESCRIPTION

This function is used to create the messages to be written or read from the Nano-ACE in a raw format

RETURN VALUE

1, 2, 3	The number of raw bytes written to/read from the device
NA_FAILURE	There was an error communicating with the Nano-ACE.
NA_INVALID_ARGUMENT	One of the arguments passed into the function is incorrect.
NA_NULL_POINTER	The array was not configured properly.
NA_NOT_ALL_BYTES_SENT	Total number bytes transmitted were less than requested.

naSimpleRead

Creates a simple message to be read from the Nano-ACE

PROTOTYPE

```
#include "na.h"
```

```
int16_t naSimpleRead(uint8_t u8MemReg, uint16_t u16Address, uint16_t *pWord);
```

PARAMETERS

u8MemReg	(input parameter) Defines either a memory or register transaction. Valid values: NA_MEMORY NA_REGISTER
u16Address	(input parameter) The Address of the Memory or Register.
pWord	(output parameter) The pointer to the data word to be read from the Nano-ACE.

DESCRIPTION

This function is used to read data back from the Nano-ACE.

RETURN VALUE

1	One data word was read.
NA_FAILURE	There was an error communicating with the Nano-ACE.
NA_INVALID_ARGUMENT	One of the arguments passed into the function is incorrect.
NA_INVALID_ADDRESS	Address provided is illegal.
NA_NOT_CONFIGURED	The SPI interface was not configured properly.
NA_SLAVE_NOT_SELECTED	The Slave Select line was not configured.
NA_NOT_ALL_BYTES_SENT	Total number bytes transmitted were less than requested.
NA_PROTOCOL_MISMATCH	Invalid protocol.
NA_NULL_POINTER	The array was not configured properly.

naSimpleWrite

Creates a Simple message to be written to the Nano-ACE

PROTOTYPE

```
#include "na.h"
```

```
int16_t naSimpleWrite(uint8_t u8MemReg, uint16_t u16Address, uint16_t u16Word);
```

PARAMETERS

u8MemReg	(input parameter) Defines either a memory or register transaction. Valid values: NA_MEMORY NA_REGISTER
u16Address	(input parameter) The Address of the Memory or Register.
u16Word	(input parameter) The data word to be written to the Nano-ACE.

DESCRIPTION

This function is used to create a Simple message to be written to the Nano-ACE.

RETURN VALUE

1	One data word was written.
NA_FAILURE	There was an error communicating with the Nano-ACE.
NA_INVALID_ARGUMENT	One of the arguments passed into the function is incorrect.
NA_INVALID_ADDRESS	Address provided is illegal.
NA_NOT_CONFIGURED	The SPI interface was not configured properly.
NA_SLAVE_NOT_SELECTED	The Slave Select line was not configured.
NA_NOT_ALL_BYTES_SENT	Total number bytes transmitted were less than requested.
NA_PROTOCOL_MISMATCH	Invalid protocol.
NA_NULL_POINTER	The array was not configured properly.

naSingleRead

Creates a Single message to be read from the Nano-ACE

PROTOTYPE

```
#include "na.h"
```

```
int16_t naSingleRead(uint8_t u8MemReg, uint16_t u16Address, uint16_t *pWord);
```

PARAMETERS

u8MemReg	(input parameter) Defines either a memory or register transaction. Valid values: NA_MEMORY NA_REGISTER
u16Address	(input parameter) The Address of the Memory or Register.
pWord	(output parameter) The pointer to the data word to be read from the Nano-ACE.

DESCRIPTION

This function is used to create a Single message to be read from the Nano-ACE.

RETURN VALUE

1	One data word was read.
NA_FAILURE	There was an error communicating with the Nano-ACE.
NA_INVALID_ARGUMENT	One of the arguments passed into the function is incorrect.
NA_INVALID_ADDRESS	Address provided is illegal.
NA_NOT_CONFIGURED	The SPI interface was not configured properly.
NA_SLAVE_NOT_SELECTED	The Slave Select line was not configured.
NA_NOT_ALL_BYTES_SENT	Total number bytes transmitted were less than requested.
NA_PROTOCOL_MISMATCH	Invalid protocol.
NA_NULL_POINTER	The array was not configured properly.

naSingleWrite

Creates a Single message to be written to the Nano-ACE

PROTOTYPE

```
#include "na.h"
```

```
int16_t naSingleWrite(uint8_t u8MemReg, uint16_t u16Address, uint16_t u16Word);
```

PARAMETERS

u8MemReg	(input parameter) Defines either a memory or register transaction. Valid values: NA_MEMORY NA_REGISTER
u16Address	(input parameter) The Address of the Memory or Register.
u16Word	(input parameter) The pointer to the data word to be written to the Nano-ACE.

DESCRIPTION

This function is used to create a Single message to be written to the Nano-ACE.

RETURN VALUE

1	One data word was written.
NA_FAILURE	There was an error communicating with the Nano-ACE.
NA_INVALID_ARGUMENT	One of the arguments passed into the function is incorrect.
NA_INVALID_ADDRESS	Address provided is illegal.
NA_NOT_CONFIGURED	The SPI interface was not configured properly.
NA_SLAVE_NOT_SELECTED	The Slave Select line was not configured.
NA_NOT_ALL_BYTES_SENT	Total number bytes transmitted were less than requested.
NA_PROTOCOL_MISMATCH	Invalid protocol.
NA_NULL_POINTER	The array was not configured properly.

naBurstRead

Creates a Burst message to be read from the Nano-ACE

PROTOTYPE

```
#include "na.h"
```

```
int16_t naBurstRead (uint8_t u8MemReg,
                    uint16_t u16Address,
                    uint16_t *pArray,
                    uint8_t u8WordCount);
```

PARAMETERS

u8MemReg	(input parameter) Defines either a memory or register transaction. Valid values: NA_MEMORY NA_REGISTER
u16Address	(input parameter) The Address of the Memory or Register.
pArray	(output parameter) Array containing the data words to be read from the Nano-ACE.
u8WordCount	(input parameter) Denotes the desired number of elements to be read from 0 to 255. 256 words will be read if 0 is entered for this parameter

DESCRIPTION

This function is used to create a Burst message to be read from the Nano-ACE.

RETURN VALUE

The number of data words that were read from the device.	
NA_FAILURE	There was an error communicating with the Nano-ACE.
NA_INVALID_ARGUMENT	One of the arguments passed into the function is incorrect.
NA_INVALID_ADDRESS	Address provided is illegal.
NA_NOT_CONFIGURED	The SPI interface was not configured properly.
NA_SLAVE_NOT_SELECTED	The Slave Select line was not configured.
NA_NOT_ALL_BYTES_SENT	Total number bytes transmitted were less than requested.
NA_PROTOCOL_MISMATCH	Invalid protocol.
NA_NULL_POINTER	The array was not configured properly.

naBurstWrite

Creates a Burst message to be written to the Nano-ACE

PROTOTYPE

```
#include "na.h"
```

```
int16_t naBurstWrite(uint8_t u8MemReg,
                    uint16_t u16Address,
                    uint16_t *pArray,
                    uint8_t u8WordCount);
```

PARAMETERS

u8MemReg	(input parameter) Defines either a memory or register transaction. Valid values: NA_MEMORY NA_REGISTER
u16Address	(input parameter) The Address of the Memory or Register.
pArray	(input parameter) The pointer to the data words to be written to the Nano-ACE.
u8WordCount	(input parameter) Denotes the desired number of elements to be written from 0 to 255. 256 words will be written if 0 is entered for this parameter

DESCRIPTION

This function is used to create a Burst message to be written to the Nano-ACE.

RETURN VALUE

1, 2, 3	The number of data words that were written to the device.
NA_FAILURE	There was an error communicating with the Nano-ACE.
NA_INVALID_ARGUMENT	One of the arguments passed into the function is incorrect.
NA_INVALID_ADDRESS	Address provided is illegal.
NA_NOT_CONFIGURED	The SPI interface was not configured properly.
NA_SLAVE_NOT_SELECTED	The Slave Select line was not configured.
NA_NOT_ALL_BYTES_SENT	Total number bytes transmitted were less than requested.
NA_PROTOCOL_MISMATCH	Invalid protocol.
NA_NULL_POINTER	The array was not configured properly.

naSetTransferMode

Sets the library to use the transfer type requested

PROTOTYPE

```
#include "na.h"
```

```
int16_t naSetTransferMode(uint8_t u8Mode);
```

PARAMETERS

u8Mode	(input parameter) Defines transfer modes. Valid values: NA_LEGACY_TRANSFERS NA_EXTENDED_TRANSFERS
--------	---

DESCRIPTION

This function is used to identify which protocol format should be used for control byte type that will be transfer to Nano Ace over SPI.

RETURN VALUE

NA_SUCCESS	The function completed without any errors.
NA_INVALID_ARGUMENT	One of the arguments passed into the function is incorrect.

naEepromSetRegisterAccess

Enables or disables access to Nano-ACE registers that are used to interact with the EEPROM.

PROTOTYPE

```
#include "eeprom.h"
```

```
int16_t naEepromSetRegisterAccess(uint8_t u8Enable);
```

PARAMETERS

u8Enable	(input parameter) Defines the EEPROM registers as either enabled or disabled. Valid values: NA_ENABLE NA_DISABLE
----------	--

DESCRIPTION

This function is used to enable or disable the Nano-ACE registers used to access EEPROM.

RETURN VALUE

NA_SUCCESS	The function completed without any errors.
NA_FAILURE	There was an error communicating with the Nano-ACE.
NA_INVALID_ARGUMENT	One of the arguments passed into the function is incorrect.
NA_INVALID_ADDRESS	Address provided is illegal.
NA_NOT_CONFIGURED	The SPI interface was not configured properly.
NA_SLAVE_NOT_SELECTED	The Slave Select line was not configured.
NA_NOT_ALL_BYTES_SENT	Total number bytes transmitted were less than requested.
NA_PROTOCOL_MISMATCH	Invalid protocol.
NA_NULL_POINTER	The array was not configured properly.

naEepromRead

Read data from EEPROM

PROTOTYPE

```
#include "eeprom.h"
```

```
int32_t naEepromRead(uint32_t u32Address,
                    uint16_t *pReadArray,
                    uint16_t u16WordCount);
```

PARAMETERS

u32Address	(input parameter) The starting 17-bit word address of EEPROM memory location.
pReadArray	(output parameter) Array containing 16-bit words read from EEPROM.
u16WordCount	(input parameter) Number of 16-bit words to be read. If '0' is entered, then 65,536 words will be read.

DESCRIPTION

This function is used to read data from the EEPROM.

RETURN VALUE

1, 2, 3	Number of data words read from EEPROM.
NA_FAILURE	There was an error communicating with the Nano-ACE.
NA_INVALID_ARGUMENT	One of the arguments passed into the function is incorrect.
NA_INVALID_ADDRESS	Address provided is illegal.
NA_NOT_CONFIGURED	The SPI interface was not configured properly.
NA_SLAVE_NOT_SELECTED	The Slave Select line was not configured.
NA_NOT_ALL_BYTES_SENT	Total number bytes transmitted were less than requested.
NA_PROTOCOL_MISMATCH	Invalid protocol.
NA_NULL_POINTER	The array was not configured properly.
NA_EEPROM_REGS_NOT_ENABLED	The Nano-ACE EEPROM registers were not enabled.
NA_EEPROM_MAX_RETRIES	Access attempts to EEPROM exceeded set limits.

naEepromVerify

Reads data from EEPROM and compares to a User-provided array of expected values.

PROTOTYPE

```
#include "eeprom.h"
```

```
int32_t naEepromVerify(uint32_t u32Address,
                      uint16_t *pVerifyArray
                      uint16_t u16WordCount);
```

PARAMETERS

u32Address	(input parameter) The starting 17-bit word address of EEPROM memory location.
pVerifyArray	(output parameter) Array containing 16-bit words that are expected to be found in EEPROM.
u16WordCount	(input parameter) Number of 16-bit words to be read/verified. If '0' is entered, then 65,536 words will be read/verified.

DESCRIPTION

This function is used to verify the data in the EEPROM is correct.

RETURN VALUE

1, 2, 3	Number of data words verified
NA_FAILURE	There was an error communicating with the Nano-ACE.
NA_INVALID_ARGUMENT	One of the arguments passed into the function is incorrect.
NA_INVALID_ADDRESS	Address provided is illegal.
NA_NOT_CONFIGURED	The SPI interface was not configured properly.
NA_SLAVE_NOT_SELECTED	The Slave Select line was not configured.
NA_NOT_ALL_BYTES_SENT	Total number bytes transmitted were less than requested.
NA_PROTOCOL_MISMATCH	Invalid protocol.
NA_NULL_POINTER	The array was not configured properly.
NA_EEPROM_REGS_NOT_ENABLED	The Nano-ACE EEPROM registers were not enabled.

NA_EEPROM_MAX_RETRIES

Access attempts to EEPROM exceeded set limits.

naEepromWrite

Writes data to EEPROM.

PROTOTYPE

```
#include "eeprom.h"
```

```
int32_t naEepromWrite(uint32_t u32Address,
                      uint16_t *pWriteArray,
                      uint16_t u16WordCount);
```

PARAMETERS

u32Address	(input parameter) The starting 17-bit word address of EEPROM memory location.
pWriteArray	(output parameter) Array containing 16-bit words to be written to the EEPROM.
u16WordCount	(input parameter) Number of 16-bit words to be written. If '0' is entered, then 65,536 words will be written.

DESCRIPTION

This function is used to write data to the EEPROM.

RETURN VALUE

1, 2, 3	Number of data words written to EEPROM
NA_FAILURE	There was an error communicating with the Nano-ACE.
NA_INVALID_ARGUMENT	One of the arguments passed into the function is incorrect.
NA_INVALID_ADDRESS	Address provided is illegal.
NA_NOT_CONFIGURED	The SPI interface was not configured properly.
NA_SLAVE_NOT_SELECTED	The Slave Select line was not configured.
NA_NOT_ALL_BYTES_SENT	Total number bytes transmitted were less than requested.
NA_PROTOCOL_MISMATCH	Invalid protocol.
NA_NULL_POINTER	The array was not configured properly.
NA_EEPROM_REGS_NOT_ENABLED	The Nano-ACE EEPROM registers were not enabled.

NA_EEPROM_MAX_RETRIES

Access attempts to EEPROM exceeded set limits.

NA_EEPROM_BAD_READBACK_
AFTER_WRITE

Data read bad after a write to a location is incorrect.

7 APPENDIX A

7.1 Error and Warning Messages

The following section contains a list of the error/warning macros that are returned by the functions of the Nano-Ace SDK.

NA_SUCCESS

Error Number: 0

Description: The function completed successfully.

NA_FAILURE

Error Number: -1

Description: There was an error communicating with the Nano-ACE.

NA_NULL_POINTER

Error Number: -2

Description: The array was not configured properly.

NA_INVALID_ARGUMENT

Error Number: -3

Description: One of the arguments passed into the function is incorrect.

NA_INVALID_ARRAY_SIZE

Error Number: -4

Description: The size of the array is not large enough to hold the returned data.

NA_NOT_CONFIGURED

Error Number: -100

Description: The SPI interface was not configured properly.

NA_SLAVE_NOT_SELECTED

Error Number: -101

Description: The Slave Select line was not configured.

NA_INVALID_ADDRESS

Error Number: -200

Description: Address provided is illegal.

NA_NOT_ALL_BYTES_SENT

Error Number: -201

Description: Total number bytes transmitted were less than requested.

NA_PROTOCOL_MISMATCH

Error Number: -202

Description: Invalid protocol.

NA_EEPROM_REGS_NOT_ENABLED

Error Number: -300

Description: Registers to access the EEPROM are not enabled.

NA_EEPROM_BAD_READBACK_AFTER_WRITE

Error Number: -301

Description: Data read from location does not equal value written.

NA_EEPROM_TIMEOUT

Error Number: -302

Description: A Timeout occurred when communicating with the EEPROM.

NA_EEPROM_MAX_RETRIES

Error Number: -303

Description: Reached maximum retries attempts to EEPROM.

NA_EEPROM_FLASH_ERROR_NO_ACK

Error Number: -304

Description: Flash error occurred, no acknowledgement from EEPROM.

NA_EEPROM_FLASH_ERROR_BAD_LOCATION

Error Number: -305

Description: Flash write error occurred.

8 APPENDIX B

8.1 RTpoll Sample

The RTpoll sample from the AceXtreme SDK (BU-69092S0) was modified to work with the Nano-ACE Library. This sample is currently not part of the Nano-ACE library package. The sample was built with Visual Studio 2010, and includes the same source files provided in the Nano-ACE library. The sample will look for the AceXtreme Header files and the AceXtreme dll and lib file. In order to rebuild this sample the AceXtreme SDK for Windows must be installed.

The sample will initialize the Evaluation board to an RT if it was not initialized by the EEPROM and start it polling for any new messages sent to the RT. Once a message is received the sample will read the message off the RT's stack and display it to the console window.

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